

week 6

1 Introduction to Computer Vision

Computer Vision is a field of Computer Science focused on enabling machines to process and interpret images and videos in a way similar to the human visual system. After capturing input from camera models, the data undergoes preprocessing, followed by the application of machine learning algorithms for pattern recognition and other tasks.

One key difference between human vision and a camera is that a camera adjusts focus by moving its lens forward or backward, whereas the human eye changes focus by altering the curvature of the lens and then both converts light to electrical signal. Human vision possesses important attributes like “**cognition**” and “**perception**”

2 Bayesian Inference

$$P(\text{World}|\text{Image}) = \frac{P(\text{Image}|\text{World}) \times P(\text{World})}{P(\text{Image})} \quad (1)$$

3 Description

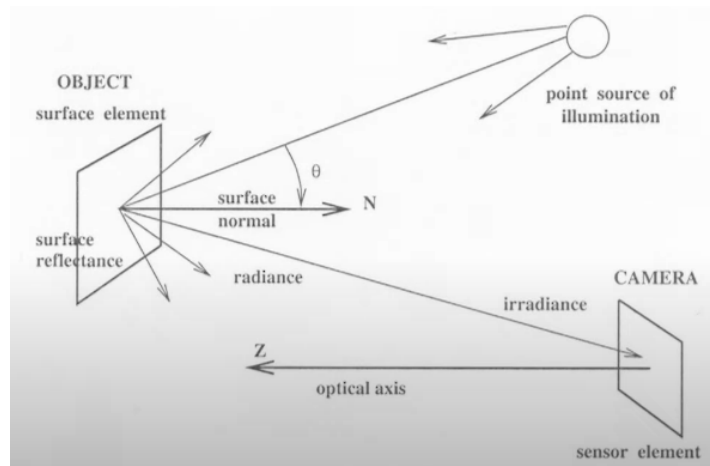
- $P(\text{World}|\text{Image})$ is the **posterior probability**: the probability of the world (the underlying state) given the observed image. In the context of computer vision, this represents inferring the real-world scene from an image.
- $P(\text{Image}|\text{World})$ is the **likelihood**: the probability of observing the image given the world. In computer graphics, this is about generating an image from a known world state.
- $P(\text{World})$ is the **prior probability**: your initial belief about the world before seeing the image.
- $P(\text{Image})$ is the **evidence** or **marginal probability**: the total probability of observing the image under all possible world states.

4 Some steps in image formation

We start by defining some terms and then the rest can be understood from the diagram.

Radiance-The amount of light energy leaving the surface element in a specific direction per unit area per unit solid angle. It describes the light traveling from the object towards the camera

Irradiance-The amount of light energy received by the camera sensor per unit area. It is the radiance integrated over all incoming directions at the sensor

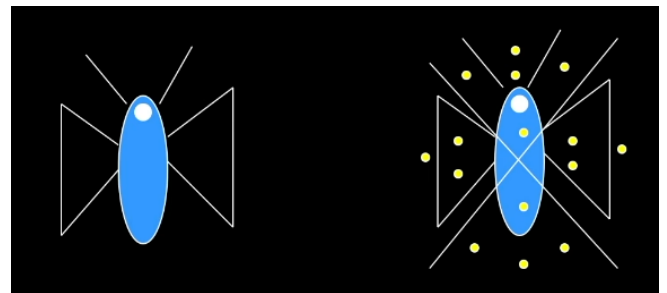
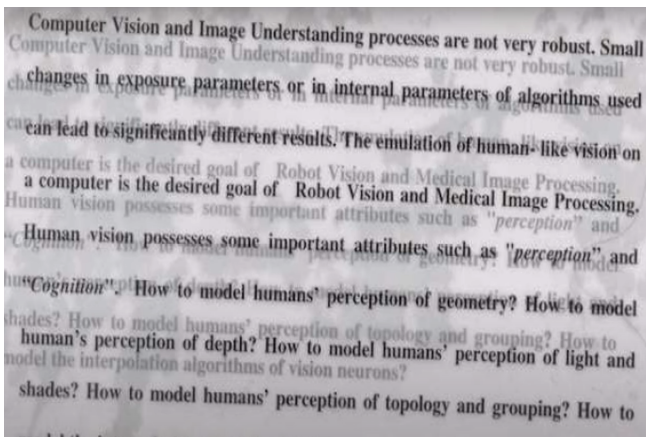


Problems difficult to tackle for computer vision

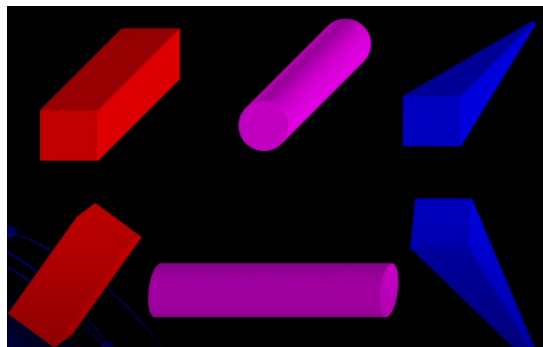
- Stereo vision for depth calculation

5 Multiple texture recognition

- Computer vision (CV) algorithms struggle with understanding the difference between an ideal (clean) image and a noisy image



- CV finds it difficult to recognize identical objects when their orientation is changed



6 Applications of computer vision

- **Facial Recognition**

Used in security systems, smartphones, and social media for identifying or verifying individuals.

- **Object Detection and Tracking**

Applied in surveillance, self-driving cars, and robotics to detect and track people, vehicles, or other objects in real-time.

- **Medical Imaging**

Helps doctors analyze X-rays, MRIs, and CT scans to detect diseases like cancer or neurological disorders.

- **Autonomous Vehicles**

Enables self-driving cars to perceive the environment—detecting lanes, traffic signs, pedestrians, and other vehicles.

- **Augmented Reality (AR)**

Enhances real-world views by overlaying digital content, used in applications like games and interior design.

- **Optical Character Recognition (OCR)**

Converts printed or handwritten text into machine-readable format, used in document scanning and translation.

7 Radiometry

Radiometry in Computer Vision refers to the measurement and modeling of light energy captured by imaging sensors. It helps quantify how much light is reflected or emitted from surfaces in a scene, which is essential for tasks like image formation, photometric analysis, and realistic rendering. Radiometric principles are used to understand brightness, shading, and contrast in images, enabling accurate interpretation and reconstruction of real-world scenes

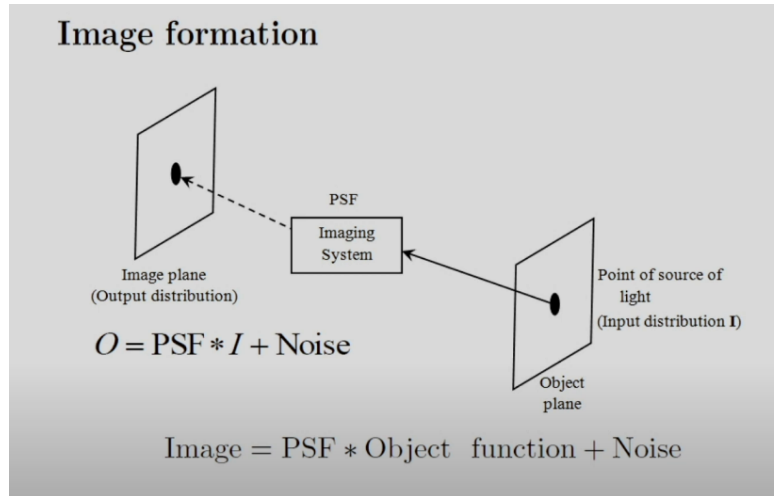
Point Spread Function

In computer vision and imaging, the **Point Spread Function (PSF)** describes how a point source of light (like a single bright dot) is represented in the image by a camera or optical system. Ideally, a point source should appear as a single pixel, but due to lens imperfections, diffraction, and sensor limitations, it gets **spread out** into a small blur. This blur is called the **Point Spread Function**. **Good cameras** have narrow PSF.

8 Noise

In image formation, it refers to **unwanted random variations** or disturbances in the image signal that **do not originate from the actual scene**. It can be caused by factors

like sensor imperfections, low light, or electronic interference, degrading image quality.



9 Brightness

- We have to measure amount of light falling on the surface and
- How the surface reflects that's light based on the reflectance property or the albedo of the surface

10 Solid angle

$$d\omega = \frac{dA}{r^2} \text{ (steradians, sr)} \quad (2)$$

- $d\omega$ is the **infinitesimal solid angle**.
- dA is the **area element** on the surface of a sphere.
- r is the **radius** of the sphere.

Hence as total surface area of area is $4\pi r^2$, hence total solid angle of sphere is 4π . One important thing to consider is **foreshortening** effect. It is an effect that occurs when an object or surface is viewed at an angle rather than head-on, causing it to appear compressed or shortened along the viewer's line of sight.

In computer vision and graphics, the foreshortening effect is mathematically described by the cosine of the angle between the surface normal and the viewing direction. The effective area of a surface patch as seen by the viewer is given by the formula:

$$A_{\text{effective}} = A \cos \theta \quad (3)$$

where A is the actual area and θ is the angle between the surface normal and the line of sight.

11 Some radiometric terms

12 Irradiance (E)

Formula:

(4)

- Irradiance is the measure of the amount of radiant flux (power, $d\phi$) received per unit area (dA) on a surface.
- It quantifies how much light energy arrives at a surface per unit area.
- Unit: Watts per square meter (W/m^2).

13 Radiance (L)

Formula:

$$L = \frac{d^2\phi}{dA \cos\theta d\Omega}$$

- Radiance measures the amount of radiant flux leaving (or arriving at) a surface per unit area (dA), per unit solid angle ($d\Omega$), and per angle (θ) between the surface normal and the direction of interest. Simply radiance per unit solid angle
- The $\cos\theta$ term accounts for the foreshortening effect when the surface is viewed at an angle.
- Unit: Watts per square meter per steradian ($\text{W}/\text{m}^2\cdot\text{sr}$).
- It is constant along straight lines in vacuum

To calculate irradiance from radiance:

$$\int_0^{2\pi} \int_0^{\pi/2} L(P, \theta_i, \phi_i) \cos\theta_i \sin\theta_i d\theta d\phi$$

$L(\mathbf{x}, \theta, \phi)$ is the radiance arriving at point $\mathbf{x}$ from direction (θ, ϕ)

Scene and Image radiance (credit YT channel "First Principles of Computer vision")

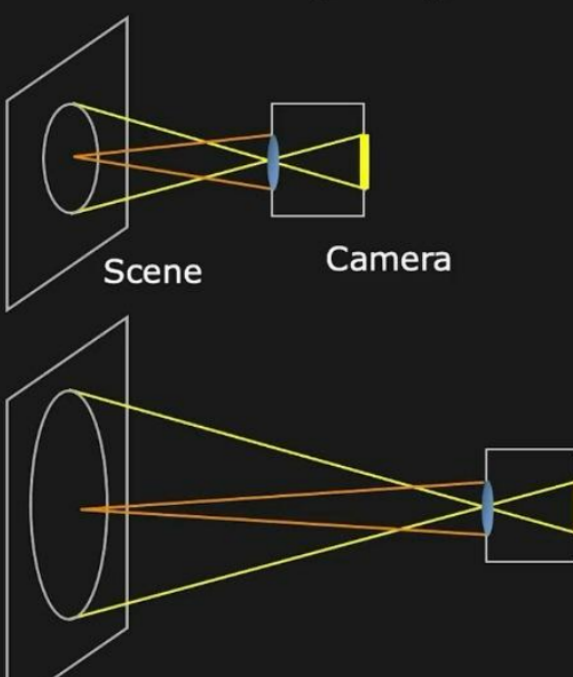
$$\mathbf{E} = \mathbf{L}(\pi/4) (\mathbf{d}/\mathbf{f})^2 \cos^4\alpha \quad (5)$$

Term	Description
E	Image Irradiance – how much light reaches the image sensor at a point.
L	Scene Radiance – how much light is being emitted or reflected by the object in the scene.
d	Aperture Diameter – diameter of the lens opening (how much light enters).
f	Focal Length – distance from lens to image plane (how focused the light is).
$(d / f)^2$	Inverse Square of the Effective F-number – controls light-gathering ability of the lens.
α (alpha)	Angle from optical axis – how far a point is from the image center.
$\cos(\alpha)$	Vignetting Factor – brightness falloff as you move away from the image center.

An interesting phenomenon is shown in the diagram below:

Scene Radiance and Image Irradiance

Does image brightness vary with scene depth? **NO**



$$E = L \frac{\pi}{4} \left(\frac{d}{f} \right)^2 \cos^4 \alpha$$

- Larger the scene depth, larger the area of light accumulation.
- Larger the scene depth, smaller the solid angle subtended by each point onto the lens, and hence less light from each point.

Bidirectional Reflectance Distribution Function

BRDF

The ratio of the radiance in the outgoing direction to the incident irradiance

$$\rho_{bd}(\theta_o, \phi_o, \theta_i, \phi_i) = \frac{dL(\theta_o, \phi_o)}{dE(\theta_i, \phi_i)}$$

$$\rho_{bd}(\theta_o, \phi_o, \theta_i, \phi_i) = \frac{L_o(P, \theta_o, \phi_o)}{L_i(P, \theta_i, \phi_i) \cos \theta_i d\omega} \text{ (sr}^{-1}\text{)}$$

All the light leaving a point is due to that arriving at that point

- Radiance leaving a surface in a particular direction:

$$L_o(P, \theta_o, \phi_o) = \rho_{bd}(\underline{x}, \theta_o, \phi_o, \theta_i, \phi_i) L_i(\underline{x}, \theta_i, \phi_i) \cos \theta_i d\omega_i$$

- Radiance leaving a surface due to its irradiance:
 - add contributions from every incoming direction

$$\int_{\Omega} \rho_{bd}(\underline{x}, \theta_o, \phi_o, \theta_i, \phi_i) L_i(\underline{x}, \theta_i, \phi_i) \cos \theta_i d\omega_i$$

These images are taken from YT channel-NPTEL IIT Guwahati